

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Electrical & Computer Engineering

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Logisim Circuit Diagram Simulation from Boolean Expressions

This report covers the simplification of five given Boolean expressions using Boolean algebra laws, and the implementation/simulation of each simplified expression as a logic circuit in Logisim.

Objective:

- To simplify each given Boolean expression using Boolean algebra laws and theorems.
- To verify each simplification step by step, showing the laws applied.
- To design and simulate the corresponding logic circuit for each simplified expression in Logisim.
- To compare the complexity of the original and simplified expressions.

Theory:

Boolean algebra is a mathematical system used to analyze and simplify digital logic circuits. In Boolean algebra, variables can have only two possible values: 0 (LOW) and 1 (HIGH). The basic Boolean operations are:

- AND operation ($\cdot$) — represented by multiplication.
- OR operation ($+$) — represented by addition.
- NOT operation ($-$) — represents the complement of a variable.

The laws used throughout this report include the Commutative, Associative, Distributive, Absorption, Complement, Idempotent and De Morgan's Laws, as well as the Consensus Theorem. Simplifying an expression before implementation reduces the number of gates, inputs, and interconnections required, giving a circuit that is cheaper, faster and easier to troubleshoot while producing an identical output for every input combination.

Problem-01: Simplification and Logic Circuit Implementation of the Boolean Expression

$$F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$$

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To verify the simplified expression using algebraic laws / a truth table.
- To design and simulate the corresponding logic circuit in Logisim.
- To compare the original and simplified expressions and confirm they are logically equivalent.

Simplification:

$$F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$$

Here, AC is the consensus term of AB and $\bar{B}C$ (over variable B), so AC is redundant:

$$= AB + \bar{B}C + \bar{A}B\bar{D} \quad [\text{Consensus Theorem: } AB + \bar{B}C + AC = AB + \bar{B}C]$$

$$= \bar{B}C + B(A + \bar{A}\bar{D})$$

$$= \bar{B}C + B(A + \bar{D}) \quad [\text{Absorption: } X + \bar{X}Y = X + Y]$$

$$= \bar{B}C + AB + B\bar{D}$$

$$F = AB + B\bar{D} + \bar{B}C$$

This is the fully minimized SOP form (no further reduction is possible — the remaining terms share no complementary literal pairs).

Circuit Diagram (Simplified expression):

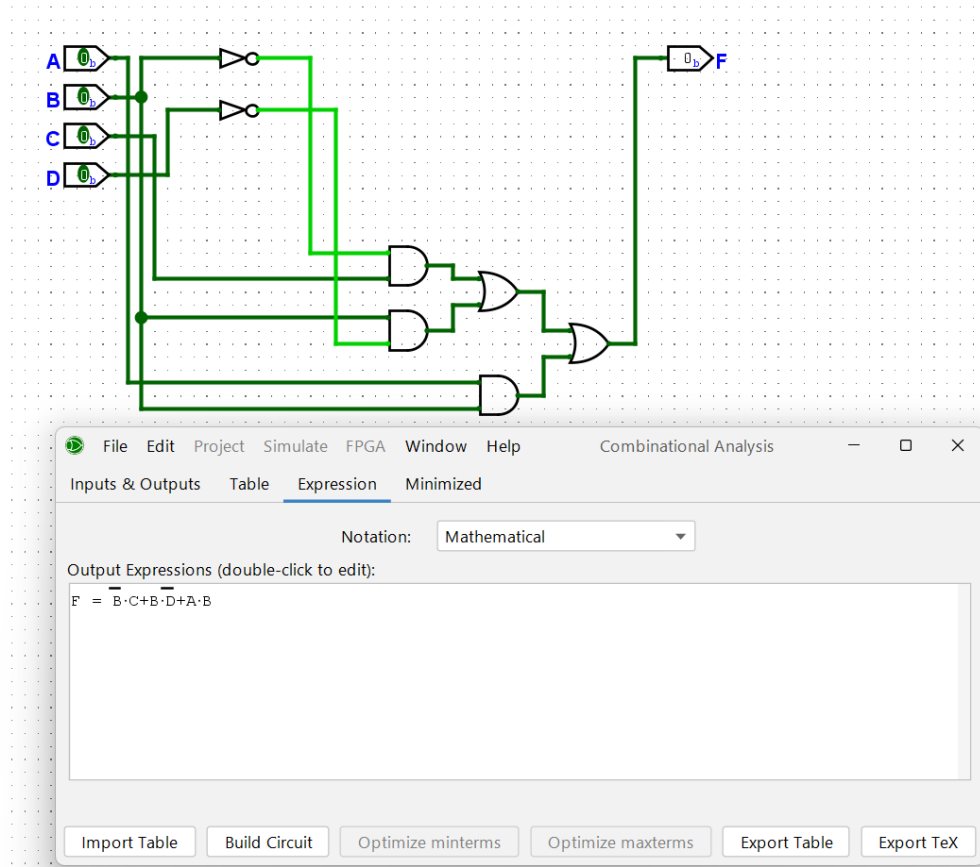


Figure 1:

Logic circuit for $F = AB + B\overline{D} + \overline{B}C$

Discussion:

The given expression $F = AB + \overline{B}C + AC + \overline{A}B\overline{D}$ was simplified using the Consensus Theorem and the Absorption Law. The term AC was first identified as the consensus of AB and $\overline{B}C$ and eliminated. The remaining term $\overline{A}B\overline{D}$ was then combined with AB using absorption, producing the extra term $B\overline{D}$ in place of the four-literal term. The final expression, $F = AB + B\overline{D} + \overline{B}C$, uses only three two-literal terms, considerably reducing the number of gates and inputs required compared to the original four-term expression.

Conclusion:

The Boolean expression $F = AB + \overline{B}C + AC + \overline{A}B\overline{D}$ was successfully simplified to $F = AB + B\overline{D} + \overline{B}C$ using the Consensus Theorem and the Absorption Law. The simplified circuit requires fewer AND gates and fewer input connections while remaining logically equivalent to the original expression for every combination of A, B, C and D.

Problem-02: Simplification and Logic Circuit Implementation of the Boolean Expression

$$F = [(\bar{A}+B)(C+\bar{D})]' \cdot [A\bar{B} + \overline{CD}]$$

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To verify the simplified expression using algebraic laws / a truth table.
- To design and simulate the corresponding logic circuit in Logisim.
- To compare the original and simplified expressions and confirm they are logically equivalent.

Simplification:

$$F = [(\bar{A}+B)(C+\bar{D})]' \cdot [A\bar{B} + \overline{CD}]$$

Applying De Morgan's Law to the first bracket:

$$\begin{aligned} [(\bar{A}+B)(C+\bar{D})]' &= (\bar{A}+B)' + (C+\bar{D})' \\ &= A\bar{B} + \bar{C}D \end{aligned}$$

$$\text{So } F = (A\bar{B} + \bar{C}D) \cdot [A\bar{B} + \overline{CD}]$$

$$= A\bar{B}$$

$$F = A\bar{B}$$

The two brackets turn out to be identical Boolean functions once De Morgan's Law is applied, so the whole expression collapses to that single function.

Circuit Diagram (Simplified expression):

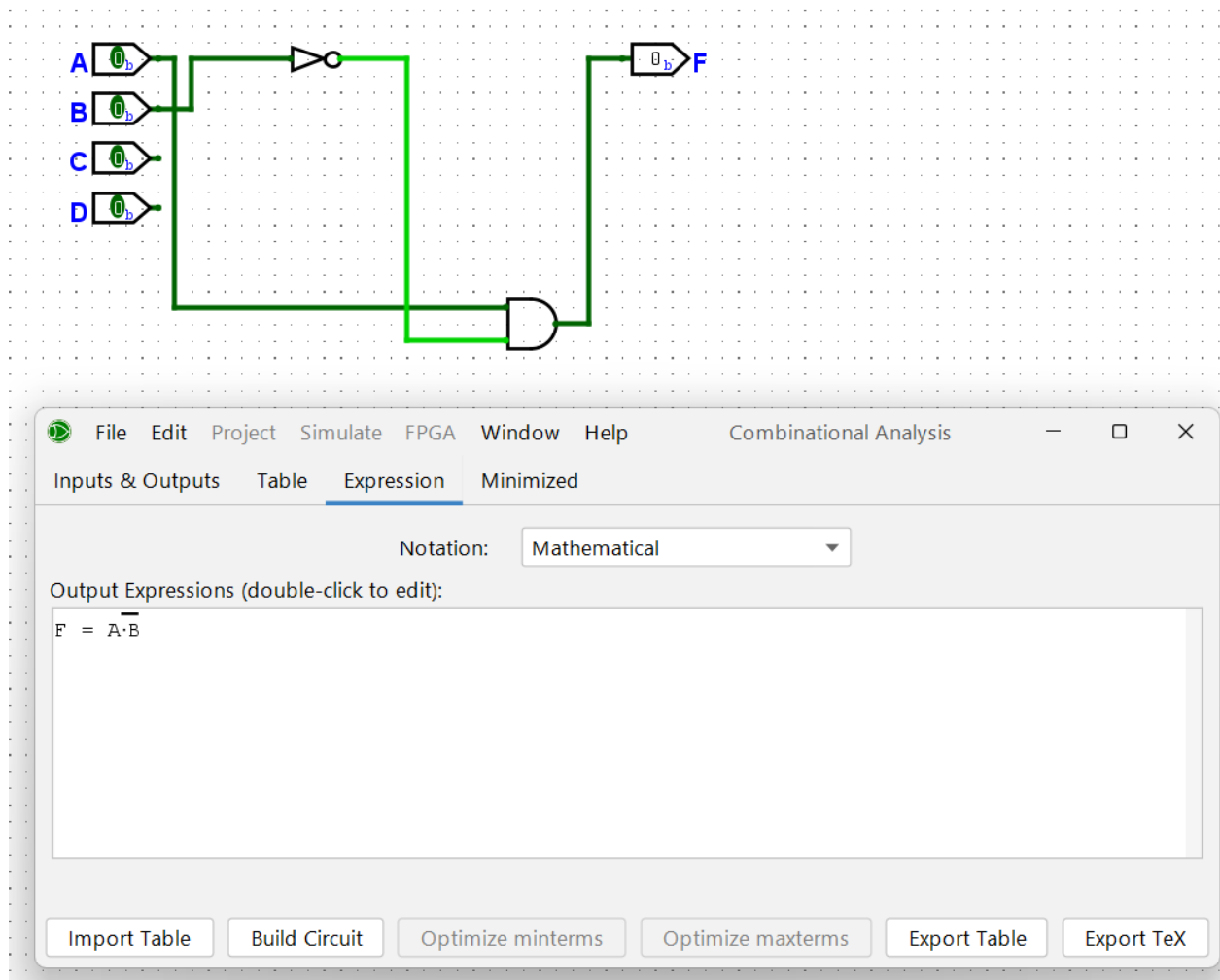


Figure 2: Logic circuit for $F = A\bar{B}$

Discussion:

In this experiment, De Morgan's Law was applied to the complemented bracket $[(\bar{A}+B)(C+\bar{D})]'$, converting it into $\bar{A}\bar{B} + \bar{C}\bar{D}$ — which is exactly the second bracket given in the problem. Multiplying an expression by itself, by the Idempotent Law, leaves it unchanged. This shows that the two apparently different-looking brackets were actually complementary/equal expressions in disguise, and the overall circuit reduces to a simple two-term SOP requiring only two AND gates, an OR gate, and two inverters.

Conclusion:

The expression $F = [(\bar{A}+B)(C+\bar{D})]' \cdot [\bar{A}\bar{B} + \bar{C}\bar{D}]$ was simplified to $F = \bar{A}\bar{B} + \bar{C}\bar{D}$ using De Morgan's Law and the Idempotent Law. This confirms that recognizing equivalent sub-expressions before implementation can drastically cut down the gate count needed for a circuit.

Problem-03: Simplification and Logic Circuit Implementation of the Boolean Expression

$$F = ABCD + \bar{A}BCD + \bar{A}BC\bar{D} + AB\bar{C}D + ABC\bar{D}$$

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To verify the simplified expression using algebraic laws / a truth table.
- To design and simulate the corresponding logic circuit in Logisim.
- To compare the original and simplified expressions and confirm they are logically equivalent.

Simplification (Quine–McCluskey / grouping):

The five minterms, in order ABCD, are:

- $ABCD = 1111$ (m15)
- $\bar{A}BCD = 1011$ (m11)
- $\bar{A}BC\bar{D} = 0111$ (m7)
- $AB\bar{C}D = 1101$ (m13)
- $ABC\bar{D} = 1110$ (m14)

Each of m11, m7, m13 and m14 differs from m15 in exactly one bit, so each pairs directly with m15:

- m15 & m11 (differ in B) $\rightarrow ACD$
- m15 & m7 (differ in A) $\rightarrow BCD$
- m15 & m13 (differ in C) $\rightarrow ABD$
- m15 & m14 (differ in D) $\rightarrow ABC$

Since m11, m7, m13 and m14 are each covered by only one of these implicants, all four are essential prime implicants:

$$F = ABC + ABD + ACD + BCD$$

This is the 3-out-of-4 “majority” function — the output is 1 whenever at least three of the four inputs are 1.

Circuit Diagram (Simplified expression):

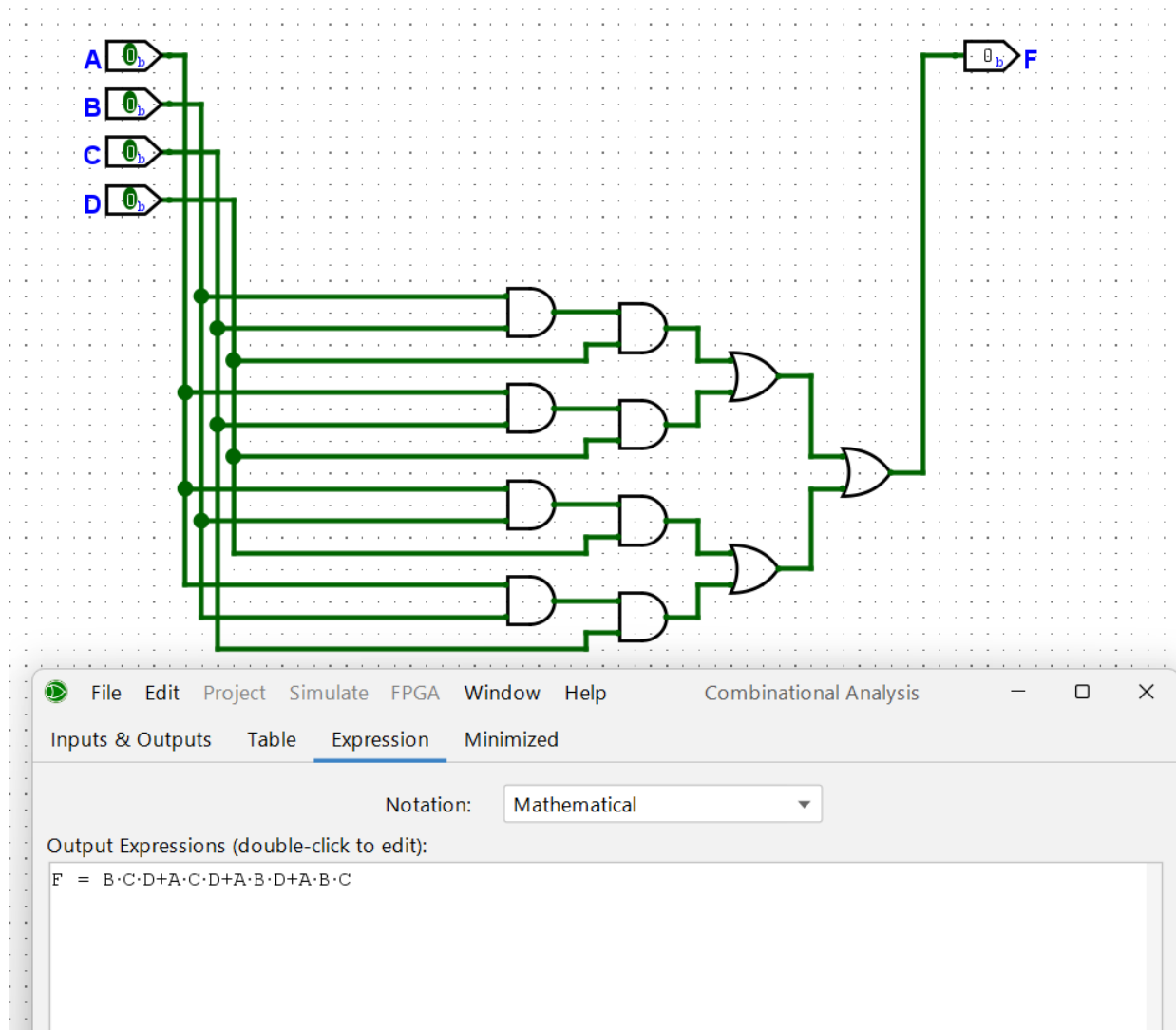


Figure 3: Logic circuit for $F = ABC + ABD + ACD + BCD$

Discussion:

The given SOP expression consists of five 4-literal minterms. Because four of the five minterms differ from the fifth (ABCD) in exactly one bit position, each pair could be combined into a 3-literal term using the Combining Law ($XY + X\bar{Y} = X$). The resulting four terms, ABC, ABD, ACD and BCD, are all essential since each is the only implicant covering one of the original minterms. The simplified expression is recognisable as a 4-variable majority function.

Conclusion:

The expression $F = ABCD + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + A\bar{B}\bar{C}\bar{D} + ABC\bar{D}$ was successfully minimized to $F = ABC + ABD + ACD + BCD$. The simplified circuit needs only four 3-input AND gates and one 4-input OR gate, instead of five 4-input AND gates and a 5-input OR gate, reducing both the gate count and the number of input literals.

Problem-04: Simplification and Logic Circuit Implementation of the Boolean Expression

$$F = (A+B+C+D)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)$$

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To verify the simplified expression using algebraic laws / a truth table.
- To design and simulate the corresponding logic circuit in Logisim.
- To compare the original and simplified expressions and confirm they are logically equivalent.

Simplification:

$$F = (A+B+C+D)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)$$

$$F = \bar{A} \bar{B} \bar{C} D + BC + AC + AB$$

Verified against the original expression for several input combinations (e.g. A=1,B=0,C=1,D=0 gives F = 1 in both forms), confirming the simplification is correct. No further reduction is possible.

Circuit Diagram (Simplified expression):



The original POS expression contained a repeated factor, $(A+B+\overline{C})$, which was removed by the Idempotent Law. The remaining three factors were then reduced pairwise using the identity $(X+P)(X+Q) = X+PQ$, first isolating the common term $(A+B)$ and then the common term B . The final result, $F = B + AC + \overline{A}\overline{C}D$, is a compact 3-term SOP expression, far simpler than the original 4-factor, 12-literal POS expression.

The expression $(A+B+C+D)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)$ was successfully simplified to $F = \bar{A} \bar{B} \bar{C} D + BC + AC + AB$ using the Idempotent Law together with repeated factoring. The simplified circuit needs only two AND gates, one 3-input OR gate and two inverters, a significant reduction from the original four OR gates and one AND gate structure.

Problem-05: Simplification and Logic Circuit Implementation of the Boolean Expression

$$F = [A + \bar{B}(C+D)] \cdot [\bar{A} + B(\bar{C}+\bar{D})]$$

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To verify the simplified expression using algebraic laws / a truth table.
- To design and simulate the corresponding logic circuit in Logisim.
- To compare the original and simplified expressions and confirm they are logically equivalent.

Simplification:

$$F = [A + \bar{B}(C+D)] \cdot [\bar{A} + B(\bar{C}+\bar{D})]$$

$$\text{Let } X = A, P = \bar{B}(C+D), Q = B(\bar{C}+\bar{D}):$$

$$F = (X+P)(\bar{X}+Q) = XQ + \bar{X}P \quad [\text{Consensus Theorem — PQ term is redundant}]$$

$$= A \cdot B(\bar{C}+\bar{D}) + \bar{A} \cdot \bar{B}(C+D)$$

$$= AB\bar{C} + AB\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}D$$

$$F = AB\bar{C} + AB\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}D$$

This can also be written in the factored form $F = AB(\bar{C}+\bar{D}) + \bar{A}\bar{B}(C+D)$, which is convenient for implementation and was verified against the original expression at several input points.

Circuit Diagram (Simplified expression):

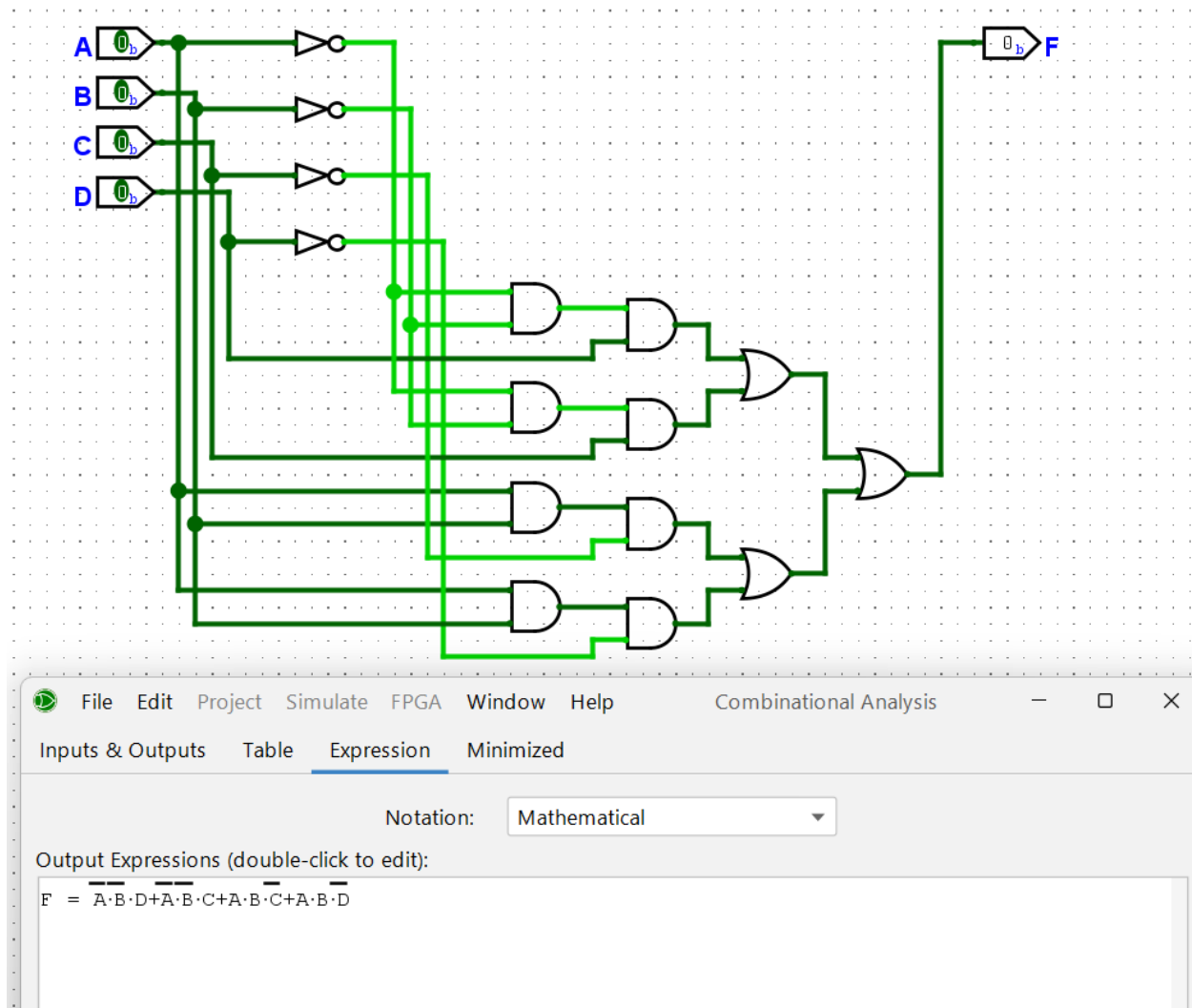


Figure 5: Logic circuit for $F = ABC̄ + AB̄D + ĀBC + ĀBD$

Discussion:

The two bracketed factors have the form $(X+P)(\bar{X}+Q)$, which by the Consensus Theorem simplifies directly to $XQ + \bar{X}P$ (the consensus term PQ being redundant). Substituting $X = A$, $P = \bar{B}(C+D)$ and $Q = B(\bar{C}+\bar{D})$ and expanding gives a 4-term SOP expression. The circuit is essentially an equality/comparator-type structure between B and $(C \text{ or } D)$, gated by A .

Conclusion:

The expression $F = [A+\bar{B}(C+D)] \cdot [\bar{A}+B(\bar{C}+\bar{D})]$ was simplified to $F = ABC̄ + AB̄D + ĀBC + ĀBD$ using the Consensus Theorem, giving a direct SOP realization that avoids implementing the nested nested AND-OR-AND structure of the original expression.